MILITARY SPECIFICATION

FLAVOXATE HYDROCHLORIDE TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Flavoxate Hydrochloride Tablets (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Flavoxate Hydrochloride Tablets, and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

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S2. Classification. Shall be type I, class 2, style A, grade 3.

S5.3 The following is added to this paragraph:

Shall contain 100 mg of flavoxate hydrochloride per tablet, within the designated assay limits for the tablets.

S6.4.9 Disintegration. Tablets shall disintegrate in not more than 15 minutes using USP simulated gastric fluid, TS, when tested by the USP method for "Uncoated Tablets".

3.1.1 Assay. The tablets shall contain not less than 93.0 percent and not more than 107.0 percent of the required amount of flavoxate hydrochloride when assayed as specified in 4.5.1.

3.1.2 Identification. The absorbance curve obtained in the assay (see 4.5.1) shall correspond to that obtained from a reference standard similarly prepared.

3.2 Ingredients.

3.2.1 Flavoxate hydrochloride.

3.2.1.1 Loss on drying. Shall not exceed 0.5 percent when an accurately weighed sample of approximately 1 gram is dried for 3 hours at 105°C.

3.2.1.2 Residue on ignition. Shall not exceed 0.2 percent using a sample of approximately 2 grams when determined by USP Method. (Reserve residue for heavy metals test.)

3.2.1.3 Heavy metals. Shall not exceed 20 ppm when determined by USP Method II.

3.2.1.4 Assays.

3.2.1.4.1 Titrimetric assay. Shall contain not less than 98.0 percent and not more than 102.0 percent of flavoxate hydrochloride when assayed as specified in 4.5.2.1.

3.2.1.4.2 Spectrophotometric assay. Shall contain not less than 98.0 percent and not more than 102.0 percent of flavoxate hydrochloride when assayed as specified in 4.5.2.2.

3.2.1.5 Identification. The absorbance spectrum of the assay solution (see 4.5.2.2) shall correspond to that obtained from a reference standard similarly prepared.

3.2.1.6 Limit of 3-methylflavone-8 carboxylic acid. Shall contain no more than 0.5 percent when determined as specified in 4.5.3.

3.3 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

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3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performacne of the inspection requirements specified herein.

4.2 Sampling. Sampling as shown in FED-STD-140 shall be applicable.

4.3 Packaging inspection. The inspection of packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

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4.5 Tests.

4.5.1 Assay of the finished tablet.

Standard preparation.

Transfer an accurately weighed sample of approximately 100 mg of flavoxate hydrochloride reference standard into a 250 ml volumetric flask. Dissolve in approximately 150 ml of 0.8 percent v/v (0.1N) hydrochloric acid by warming. Cool to room temperature and dilute to volume with 0.8 percent hydrochloric acid. Dilute 10.0 ml to 100.0 ml with 0.8 percent hydrochloric acid.

Procedure.

Finely powder not less than 20 tablets. Transfer an accurately weighed sample of the powder (equivalent to about 100 mg of flavoxate hydrochloride) to a 250 ml volumetric flask. Add about 100 ml of 0.8 percent v/v HCl (0.1N) and heat on a steam bath for 30 minutes, swirling occasionally. Cool to room temperature, dilute to volume with 0.8 percent HCl and mix well. Filter a portion of the solution, discarding the first 15-20 ml of filtrate. Dilute 10.0 ml to 100.0 ml with 0.8 percent HCl and mix well. Record the ultraviolet absorption spectrum on a suitable spectrophotometer from 380 to 220 nm using the following operating conditions: 1 cm cells, 0.8 percent HCl in the reference cell, each chart division = 2.5 nm. Record a baseline using the same operating conditions with 0.8 percent HCl in both cells.

Calculation:

$$\frac{Au \times \text{mg standard} \times \text{average tablet weight (mg)}}{As \times \text{Sample weight (mg)}} = \text{mg flavoxate hydrochloride per tablet}$$

where Au = Absorbance sample 318 nm - absorbance 370 nm
As = Absorbance standard 318 nm - absorbance 370 nm

4.5.2 Assays (active ingredient).

4.5.2.1 Titrimetric assay.

Transfer an accurately weighed sample of approximately 100 mg to a 250 ml Erlenmeyer flask and dissolve in 50 ml of glacial acetic acid. Add 10 ml of 6 percent mercuric acetate solution

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USP, TS, 2 drops of methyl violet TS, and titrate with 0.1N perchloric acid in glacial acetic acid, volumetric solution, to an emerald green endpoint. Perform a blank determination and make any ncecssary correction. Each ml of 0.1N $HClO_4$ is equivalent to 42.79 mg of flavoxate hydrochloride. Calculate the percent of flavoxate hydrochloride on the anhydrous basis.

4.5.2.2 Spectrophotometric assay.

Quantitatively transfer an accurately weighed sample of approximately 100 mg to a 250 ml volumetric flask. Dissolve in approximately 150 ml of 0.8 percent v/v (0.1N) hydrochloric acid, by warming. Cool to room temperature and dilute to volume with 0.8 percent hydrochloric acid. Dilute 10.0 ml to 100.0 ml with 0.8 percent hydrochloric acid.

Prepare a solution of flavoxate hydrochloride reference standard with a known final concentration of approximately 40 mcg per ml in 0.8 percent hydrochloric acid. Concomitantly determine absorbances of standard and sample with a suitable recording spectrophotometer from 380 to 220 nm using 1 cm cells, 0.8 percent HCl in the reference cell, and each chart division equal to 2.5 nm. Record a baseline using the same operating conditions with 0.8 percent HCl in both cells.

Calculation:

$$\frac{Au \times C \times 250 \times 100}{As \times S \times 10} \times 100 = \text{percent purity}$$

where Au = Absorbance sample 318 nm-absorbance 370 nm
As = Absorbance standard 318 nm-absorbance 370 nm
C = Concentration standard mg/ml
S = Sample Weight (mg)

4.5.3 Limit test.

Standard Solution.

Into a 100 ml volumetric flask dissolve 50 mg of 3-Methylflavone-8-Carboxylic Acid reference standard with chloroform and dilute to volume. Mix well. Transfer a 5 ml aliquot of this solution to a 100 ml volumetric flask, dilute to volume with chloroform and mix. This solution contains 0.25 mcg per 10 ul.

Procedure

Transfer a 50 mg sample of flavoxate hydrochloride to a 10 ml volumetric flask. Dissolve in and dilute to volume with chloroform and mix well. Use a micropipet to transfer 10 microliters of this solution to a point about 1.5 cm from the bottom edge of a silica gel G

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coated plate. Transfer 10 microliters of the standard solution to another point, the same distance from the bottom edge and 2 cm away from the sample. Add 0.4 ml of conc. ammonium hydroxide to 100 ml of acetone and add enough of this freshly prepared solution to a chromatographic chamber to make a layer about 1/4 inch deep. Place the silica gel plate in the chamber and allow the solvent front to rise about 7 cm above the origin. Remove the plate from the chamber and dry it in a stream of air.

Spray the chromatogram with 10 percent v/v H_2SO_4 and heat the plate in an oven at 105°C for 15 minutes. Examine the plate under long wavelength ultraviolet light. The intensity of the minor sample spot, whose R_f value corresponds with that of the standard, shall not exceed that of the standard.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 100 tablets, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Thirty-six (36) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Thirty-six (36) units of issue shall be packed in an exterior container constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

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5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)"

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

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6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-172-3420	FLAVOXATE HYDROCHLORIDE TABLETS, 100 mg, 100s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA - DM

Project No. 6505-1845

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-37700 FLAVOXATE HYDROCHLORIDE TABLETS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

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DATE

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